Geographical risk assessment from tsunami run-up events based on socioeconomic-environmental data

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Abstract

This study proposes a method to measure geographical risk from tsunami run-up events based on socioeconomic-environmental data. The physical exposure of Japan to tsunami run-up events is computed based upon 1-km grid square statistics from the population census of the Japanese Statistics Bureau of the Ministry of Internal Affairs and Communication, 1-km grid square national land numerical information downloaded from the National and Regional Policy Bureau of the Japanese Ministry of Land, Infrastructure, Transport and Tourism, and tsunami run-up catalog data downloaded from NOAA Tsunami Data and Information. Using estimates of physical exposure in each 1-km square grid, we define the physical exposure of potential passengers and freight as the sum of physical exposures over the area close to an airport. Information on Japanese domestic air transportation is further extracted as a network consisting of airports and flights. Using an exposed value as the sum of passengers of flights landing at and taking off from each airport and the sum of freight weight at each airport, we compute values of the physical exposure of both confluence and logistics for 86 Japanese airports in terms of tsunami run-up events.

Keyword: generalized Pareto distribution, risk, hazard, vulnerability

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